Fatemeh Fardno

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Research Interests

Game theory, Reinforcement learning, Online learning

Education

University of Illinois Urbana-Champaign Champaign, IL, USA
PhD in Electrical and Computer Engineering Jan 2024 – Present
Advisor: Seyed Rasoul Etesami. *GPA: 4.0.*

University of Waterloo Waterloo, ON, Canada
MASc in Electrical and Computer Engineering Sep 2020 – Sep 2022
Advisor: Seyed Majid Zahedi. *GPA: 3.8.*

Sharif University of Technology Tehran, Iran
BS in Electrical Engineering Sep 2015 – Dec 2019
Advisor: Mohammad Saleh Tavazoei. *GPA: 3.87.*

Honors and Awards

James W. Beauchamp Award for Interdisciplinary Research 2026
University of Illinois Urbana-Champaign

Best Poster Award 2026
21st CSL Student Conference

International Master's Award of Excellence 2020–2022
University of Waterloo

Membership of National Elite Foundation 2015–2019
Sharif University of Technology

3rd Prize in the 15th Austrian Physicists' Tournament (AYPT) 2013
1st Prize in the 5th Persian Physicists' Tournament (PYPT) 2012

Publications

Dynamic Transaction Scheduling and Pricing in the Ethereum Mempool
Fatemeh Fardno, S. Rasoul Etesami.
Under Review.

A game-theoretic framework for distributed load balancing: Static and dynamic game models
Fatemeh Fardno, S. Rasoul Etesami.
Automatica, Volume 185, March 2026.

Beyond Theorems: A Counterexample to Potential Markov Game Criteria
Fatemeh Fardno, S. Majid Zahedi.
Under Review.

Power allocation of sensor transmission for remote estimation over an unknown Gilbert-Elliott channel
Tahmoores Farjam, Fatemeh Fardno, Themistoklis Charalambous.
2020 European Control Conference (ECC).

Research Experience

Graduate Research Assistant at the Coordinated Science Laboratory (CSL), UIUC
Mentor: Professor S. Rasoul Etesami — Jan 2024 – Present

Graduate Research Assistant at the Multi-agent Systems Lab (MASLAB), University of Waterloo
Mentor: Professor S. Majid Zahedi — Sep 2020 – Sep 2022
Conducting research at the intersection of multi-agent reinforcement learning and game theory, with applications to distributed load balancing.
Thesis title: Fictitious Mean-field Reinforcement Learning for Distributed Load Balancing

Research Intern at the Distributed Systems Control Group, Aalto University
Mentor: Professor Themistoklis Charalambous — Jun 2019 – Sep 2019
Designed a sensor scheduling policy for remote state estimation over time-varying communication channels. Modeled the system as a partially observable Markov decision process (POMDP) and applied Bayesian inference to learn channel statistics.

Research Assistant at the Superconductor Electronics Research Laboratory (SERL), Sharif University of Technology
Mentor: Professor Mehdi Fardmanesh — Sep 2017 – Sep 2018
Fabricated microfluidic chips for high-sensitivity electrical measurements.

Teaching Experience

Teaching assistant, ECE, UIUC — Fall 2024 – Present
ECE 110: Introduction to Electronics
ECE 313: Probability with Engineering Applications
ECE 314: Probability in Engineering Lab

Teaching assistant, ECE, University of Waterloo — Fall 2021 – Spring 2022
ECE 108: Discrete Mathematics and Logic

Teaching assistant, ECE, Sharif University of Technology — Fall 2019

Engineering Statistics and Probability

Industry Experience

Microchip Technology Inc., FPGA Business Unit — Toronto, ON, Canada
Software Engineer — Oct 2022 – Dec 2023
Full-time software engineer in the Timing, Power and Device Data Group. Built and deployed an XGBoost-based delay prediction model for post-synthesis netlists. Improved internal tooling through code reviews, unit tests, and software quality enforcement. Collaborated cross-functionally with hardware and verification teams.

Dana Control — Tehran, Iran
Control Engineering Intern — Summer 2018
Programmed PLC systems and designed HMI interfaces for industrial automation.

Poster Presentations

A Game-Theoretic Framework for Distributed Load Balancing: Static and Dynamic Game Models
CSL Student Conference, February 2026
Allerton Conference, September 2025

Skills

Programming
Proficient in Python, C/C++, SQL, MATLAB, and Verilog.

Languages
Farsi, English, Turkish

Service

UIUC ECE GradSAC — Sep 2024 – Present
Represent graduate student interests in departmental discussions. Support organization of graduate student events and community-building activities.

Reviewer — Fall 2025 – Present
Automatica, IEEE Conference on Decision and Control (CDC)